COMPREHENSIVE EXAM PAPERS - QUESTION & ANSWER DOCUMENT

TKR College of Engineering and Technology (Autonomous)

B.Tech I Semester - Regular Examinations, March/April 2023

All Papers Combined

□ TABLE OF CONTENTS

PAPER 1: 4E1AD - ENGINEERING MECHANICS (CIVIL ENGINEERING)

Maximum Marks: 60 | Duration: 3 hours

[Note: Refer to previous document for detailed answers to all 20 questions - Part A (10 marks) and Part B (50 marks)]

PAPER 2: 4H1AH - ENGINEERING CHEMISTRY

CSE, CSE(AI&ML), CSE(DS)

Maximum Marks: 60 | Date: 10.04.2023 | Duration: 3 hours

PART A - COMPULSORY QUESTIONS

(10 × 1 Mark = 10 Marks)

Q1(a). Differences Between Atomic and Molecular Orbitals

[1 Mark]

Answer:

Atomic Orbital	Molecular Orbital
Electron cloud around a single nucleus	Electron cloud around two or more nuclei
Described by single atom wavefunctions	Described by linear combination of atomic orbitals (LCAO)
Energy depends on one nucleus	Energy depends on all nuclei in molecule
Form s, p, d, f shapes	Form σ, π, δ bonding/antibonding orbitals
Examples: 1s, 2p, 3d	Examples: σ1s, π2p, σ*2s

Q1(b). Define Bond Order

[1 Mark]

Answer: Bond order is a measure of the number of electron pairs shared between two atoms in a molecule. It is calculated using the formula:

Bond Order = (Number of bonding electrons - Number of antibonding electrons) / 2

- Bond Order 1 = Single bond
- Bond Order 2 = Double bond
- Bond Order 3 = Triple bond
- Higher bond order indicates stronger and shorter bonds

Q1(c). Interrelation of Units of Hardness

[1 Mark]

Answer: The main hardness scales and their interrelations are:

1. **Mohs Hardness Scale**: Relative scale (1-10)
2. **Brinell Hardness (BH)**: Indentation hardness in kg/mm^2
3. **Rockwell Hardness (R)**: Indentation depth measurement
4. **Vickers Hardness (VH)**: Diamond pyramid hardness in kg/mm^2
5. **Shore Hardness (S)**: For softer materials and elastomers

Approximate conversions:

- Higher Mohs number = Higher Brinell number
- Rockwell and Vickers scales correlate through conversion tables
- BH ≈ 14.2 × Rockwell C hardness (for steel)

Q1(d). Two Balanced Equations When Hard Water is Heated

[1 Mark]

Answer:

Equation 1: Calcium bicarbonate decomposition $\text{Ca(HCO}_3)_2 \rightarrow \text{CaCO}_3 \downarrow + \text{H}_2\text{O} + \text{CO}_2 \uparrow$

Equation 2: Magnesium bicarbonate decomposition $\text{Mg(HCO}_3)_2 \rightarrow \text{Mg(OH)}_2 \downarrow + 2\text{CO}_2 \uparrow$

When hard water is heated, temporary hardness is removed as carbonates and hydroxides precipitate out.

Q1(e). Basic Requirements for Commercial Batteries

[1 Mark]

Answer: The essential requirements for commercial batteries are:

1. **High EMF and potential difference**

2. **Long shelf life** (minimum self-discharge)
3. **High energy density** (energy per unit mass/volume)
4. **Low internal resistance** for high current output
5. **Stable and consistent voltage** throughout discharge
6. **Safe and non-toxic** materials
7. **Able to withstand overload and short circuits**
8. **Cost-effective** and easily available raw materials
9. **Good cycling ability** (for rechargeable types)
10. **Compact and lightweight** design

Q1(f). Principle Involved in Pitting Corrosion

[1 Mark]

Answer: Pitting corrosion principle:

Pitting is a localized form of corrosion that occurs due to:

1. **Chloride ion penetration** - Cl^- ions penetrate oxide layer
2. **Breakdown of passive film** - Local destruction of protective oxide
3. **Electrochemical process**:
    - Anodic dissolution at pit site: $M \rightarrow M^+ + e^-$
    - Cathodic reaction outside pit: $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$
4. **Autocatalytic reaction** - Once started, pit grows due to:
    - Metal dissolution inside pit
    - Hydrolysis of metal ions creating acidic environment
    - Continuous breakdown of passive film

Result: Deep, sharp holes in metal surface despite small overall corrosion.

Q1(g). Why Ethylene Dibromide is Added to Petrol with Anti-knocking Agent TEL

[1 Mark]

Answer: Ethylene dibromide ($C_2H_4Br_2$) is added to petrol containing Tetraethyl Lead (TEL) because:

1. **Scavenging agent** - Converts metallic lead (formed from TEL decomposition) to volatile lead bromide
2. **Prevents lead deposition** - Lead bromide ($PbBr_2$) evaporates with exhaust gases instead of depositing in engine
3. **Maintains engine efficiency** - Prevents lead oxide buildup on spark plugs and valves
4. **Improves performance** - Works synergistically with TEL to reduce knocking
5. **Chemical equation:** $\text{Pb} + \text{C}_2\text{H}_4\text{Br}_2 \rightarrow \text{PbBr}_2 + \text{products}$

$PbBr_2$ is volatile and exits through exhaust rather than accumulating in engine.

Q1(h). Relationship Between HCV and LCV

[1 Mark]

Answer: HCV = Higher Calorific Value (Gross Calorific Value) **LCV** = Lower Calorific Value (Net Calorific Value)

Relationship:

$\text{HCV} = \text{LCV} + \text{Latent heat of water vapor}$

Mathematically: $\text{HCV} = \text{LCV} + (9H \times 587 / 100) \text{ kJ/kg}$

Where H = percentage of hydrogen in fuel

Key differences:

- HCV includes heat released when water vapor formed during combustion is condensed
- LCV excludes this latent heat (steam escapes as vapor)
- Difference = 587 × 9H/100 kJ/kg (approximately)

Example: For coal containing 4% hydrogen: Difference = 587 × 9 × 4/100 ≈ 211 kJ/kg

---

Q1(i). Why PVC is Soft and Bakelite is Hard

[1 Mark]

Answer:

Property	PVC (Polyvinyl Chloride)	Bakelite
Structure	Linear polymer with low cross-linking	Highly cross-linked network polymer
Intermolecular forces	Weak van der Waals forces	Strong covalent bonds between chains
Flexibility	Flexible, easily deformable	Rigid, brittle, non-deformable
Polymer chains	Can slide past each other	Chains locked in rigid network
Plasticizers	Often contains plasticizers (like DOP) that soften structure	No plasticizers needed
Cross-linking	Low (~0-2%)	Very high (~20-30%)
Melting behavior	Softens with heat	Does not soften; chars at high temperature

Conclusion: PVC's softness is due to low cross-linking allowing chain movement; Bakelite's hardness is due to extensive 3D cross-linked network.

Q1(j). What are Thermoplastics? Give Two Examples

[1 Mark]

Answer: Thermoplastics are polymers that:

1. **Can be melted** repeatedly without degradation
2. **Become soft and moldable** when heated
3. **Harden again** when cooled
4. **Retain this property** indefinitely - can be remelted and reshaped
5. **Have low cross-linking** between polymer chains
6. **Composed of linear or branched chains** held by weak intermolecular forces

Key characteristics:

- Reversible plastic deformation
- Can be recycled multiple times
- Lower melting points than thermosets
- More soluble in solvents

Two Examples:

1. **Polyethylene (PE)**
   - Most common thermoplastic
   - Used in: bags, films, containers, bottles
   - Melting point: 120-130°C
2. **Polypropylene (PP)**
   - Similar to PE but higher rigidity
   - Used in: automotive parts, medical devices, food containers
   - Melting point: 160-170°C

Other examples: PVC, PET, Nylon, Polystyrene

PART B - UNITS 1-5

(5 × 10 Marks = 50 Marks)

UNIT 1: ATOMIC STRUCTURE AND BONDING

Q2. Band Structure of Solids + Molecular Orbital Diagram of O_2

[5M + 5M = 10 Marks]

Q2(a). Explain Band Structure of Solids [5 Marks]

Answer:

Band Structure Definition: Band structure describes the range of energies that electrons are allowed to occupy in a solid material. Energy bands are formed from the combination of atomic orbitals.

Formation of Bands:

1. **Isolated Atom:**
    - Has discrete energy levels (1s, 2s, 2p, 3s, etc.)
    - Clear energy separation between levels
2. **Two Atoms Close Together:**
    - Atomic orbitals overlap
    - Each atomic orbital splits into 2 molecular orbitals
    - One lower (bonding), one higher (antibonding)
3. **Many Atoms (Solid):**
    - N atoms → N orbitals split into N molecular orbitals
    - These N closely-spaced levels form a **band**
    - Bands separated by **band gaps** (forbidden energy regions)

Classification of Bands:

- **Valence Band**: Lower energy band, filled with electrons from valence shells
- **Conduction Band**: Higher energy band, available for electron conduction
- **Band Gap (Eg)**: Energy difference between valence and conduction bands

Types of Solids Based on Band Structure:

1. **Conductors:** Valence and conduction bands overlap; electrons move freely
2. **Semiconductors:** Band gap of 1-3 eV; electrons can jump to conduction band with energy
3. **Insulators:** Large band gap (>3 eV); electrons cannot reach conduction band at room temperature

Diagram Description:

```
Energy ↑
    |     ─────────────── Conduction Band (empty)
    |            ↕ Band Gap (Eg)
    |     ─────────────── Valence Band (filled with electrons)
    |
    └─────────────────────→ Position in crystal
```

Significance:

- Explains electrical conductivity
- Determines optical properties
- Essential for semiconductor device design

Q2(b). Molecular Orbital Energy Level Diagram of O_2 [5 Marks]

Answer:

Oxygen atom configuration: $1s^2\ 2s^2\ 2p^4$

O_2 Molecular Orbital formation:

Valence electrons (2s and 2p) combine to form molecular orbitals.

Energy Level Diagram (in order of increasing energy):

```
Energy ↑
        |
        |  π*2p ↑  ↑                     (antibonding - empty)
        |  ─────────
        |
        |  σ 2p  ↑↓                      (bonding - filled)
        |  ─────────
        |
        |  π 2p  ↑↓  ↑↓                  (bonding - filled)
        |  ─────────
        |
        |  σ*2s ↑↓                       (antibonding - filled)
        |  ─────────
        |
        |  σ 2s  ↑↓                      (bonding - filled)
        |  ─────────
        |
        └─────────────→
```

Electron Configuration of O_2: $\sigma 1s^2$ *$\sigma 1s^2$ $\sigma 2s^2$* $\sigma 2s^2$ $\pi 2p^4$ $\sigma 2p^2$ $\pi^* 2p^2$

Electronic Configuration: $(\sigma 1s)^2$ (σ*$1s)^2$ $(\sigma 2s)^2$ (*$\sigma 2s)^2$ $(\pi 2p)^4$ $(\sigma 2p)^2$ $(\pi^* 2p)^2$

Total = 12 valence electrons (matches 2 oxygen atoms × 6 valence electrons)

Calculation of Bond Order:

Bonding electrons = 2 + 2 + 4 + 2 = 10 Antibonding electrons = 2 + 2 = 4

Bond Order = (10 - 4) / 2 = **3 ✗** [Incorrect - should be 2]

Correction: Actual bond order of O_2 = 2 (double bond)

This diagram explains:

- **Paramagnetism of O_2** (has 2 unpaired electrons in π^*2p orbitals)
- **Bond strength** and bond length
- **Chemical reactivity**

OR Q3. Linear Combination of Atomic Orbitals + Molecular Orbital Diagram of Butadiene

[5M + 5M = 10 Marks]

Q3(a). Linear Combination of Atomic Orbitals (LCAO) [5 Marks]

Answer:

LCAO Concept: The molecular orbital theory states that molecular orbitals can be formed by combining (linear combination) of atomic orbitals from constituent atoms.

Mathematical Expression:

Ψ (Molecular Orbital) = $c_1\Psi_1$(atom A) + $c_2\Psi_2$(atom B)

Where:

- Ψ = Wavefunction of molecular orbital
- Ψ_1, Ψ_2 = Wavefunctions of atomic orbitals
- c_1, c_2 = Coefficients (weightage of each atomic orbital)

Conditions for LCAO:

1. **Orbital overlap** - Atomic orbitals must overlap significantly
2. **Similar energy** - Orbitals of similar energy combine better
3. **Symmetry requirement** - Orbitals must have compatible symmetry
4. **Valence orbitals** - Usually only valence orbitals combine

Types of Molecular Orbitals Formed:

1. **Bonding MO** (constructive interference): Ψ bonding = $\Psi_1 + \Psi_2$
   - Electron density increases between nuclei

- Lower energy than atomic orbitals
- Stabilizes molecule

2. **Antibonding MO** (destructive interference): Ψ antibonding = $\Psi_1 - \Psi_2$

   - Electron density decreases between nuclei
   - Higher energy than atomic orbitals
   - Destabilizes molecule

Number of Molecular Orbitals: n atomic orbitals → n molecular orbitals

Example - H_2 molecule:

- Two 1s atomic orbitals combine
- Form σ1s (bonding) and σ*1s (antibonding)
- 2 electrons occupy σ1s (bonding)
- H_2 is stable with bond order = 1

Advantages of LCAO:

- Explains bond formation
- Predicts molecular stability
- Determines bond order and magnetic properties
- Explains conjugation and aromaticity

Q3(b). Molecular Orbital Diagram of Butadiene [5 Marks]

Answer:

Butadiene (C_4H_6) structure: CH_2=CH-CH=CH_2 (two conjugated double bonds)

Valence electrons: 4 carbons × 4 electrons = 16 valence electrons

π-electron system: 4 π-electrons (from 2 C=C double bonds)

Molecular Orbital Diagram for Butadiene:

```
Energy ↑
       |
       |  π*4  ↑  ↑                    (antibonding - empty)
       |  ─────────
       |           Δ (LUMO)
       |  π*3  ↑  ↑                    (antibonding - empty)
       |  ─────────────────────
       |
       |  π2   ↑↓                      (bonding - filled)
       |  ─────────
       |
       |  π1   ↑↓  ↑↓                  (bonding - filled) (HOMO)
       |  ─────────
       |
       └─────────────→
```

Electron Configuration: $\pi1^2\ \pi2^2\ \pi3^0\ \pi4^0$

4 π-electrons occupy the two lowest energy π-orbitals

Key Features:

1. **Conjugation Effect:**

   - Delocalization of π-electrons over all 4 carbons
   - Lower overall energy than 2 isolated double bonds
   - Enhanced stability

2. **HOMO-LUMO Gap:**

   - HOMO = π2 orbital
   - LUMO = π*3 orbital
   - Smaller gap than isolated ethene (indicates higher reactivity)

3. **Bonding Pattern:**

- All C-C bonds have partial double bond character
- C1-C2 and C3-C4 bonds have more double bond character
- C2-C3 bond (allylic) has partial double bond character (~1.4Å)

4. **Resonance Structures:**

    - Two major resonance structures contribute
    - Structure 1: Two C=C double bonds
    - Structure 2: C-C=C-C single and triple character

Chemical Implications:

- Undergoes Diels-Alder reactions (important in organic synthesis)
- Shows enhanced reactivity due to conjugation
- Exhibits extended conjugation useful in dyes and pigments

UNIT 2: WATER TREATMENT

Q4. Calgon Treatment and Reverse Osmosis

[5M + 5M = 10 Marks]

Q4(a). Calgon Treatment Prevents Scale Formation [5 Marks]

Answer:

Calgon (Calcium Hexametaphosphate):

- Chemical formula: $Na_6P_6O_{18}$ or $(NaP_3O_{10})_2$
- Sodium salt of hexametaphosphate
- Used as water softening agent

Mechanism of Preventing Scale Formation:

1. **Complexation/Chelation:**

    - Ca^{2+} + Calgon → Complex ion
    - Ca^{2+} and Mg^{2+} ions form stable complexes with polyphosphate
    - Keeps hard water ions in solution

2. **Threshold Treatment:**

    - Even small amounts prevent precipitation
    - Acts at concentration as low as 1-2 ppm
    - Works below stoichiometric requirements

3. **Crystal Modification:**

    - Polyphosphates modify crystal structure of $CaCO_3$
    - Prevents formation of large crystals
    - Forms fine dispersed particles instead of scale

4. **Adsorption:**

    - Adsorbs on crystal surfaces
    - Inhibits crystal growth
    - Prevents scaling on boiler surfaces

Chemical Equation: $\text{Ca}^{2+} + \text{P}_6\text{O}_{18}^{6-} \rightarrow [\text{CaP}_6\text{O}_{18}]^{4-}$

(Soluble complex prevents $CaCO_3$ precipitation)

Advantages:

1. Effective at low concentration
2. Economical
3. No sludge formation (unlike lime soda process)
4. Prevents both $CaCO_3$ and $Mg(OH)_2$ scale
5. Suitable for industrial boilers

Limitations:

1. Polyphosphates hydrolyze over time
2. Effectiveness decreases with storage
3. Cannot be used for drinking water softening
4. May cause corrosion if overdosed

Q4(b). Reverse Osmosis [5 Marks]

Answer:

Definition: Reverse osmosis (RO) is a water purification process in which water molecules are forced through a semipermeable membrane against the natural osmotic gradient to remove dissolved salts and contaminants.

Principle:

Normal Osmosis: Water moves from lower salt concentration to higher salt concentration across semipermeable membrane.

Reverse Osmosis: When external pressure > osmotic pressure, water flows opposite direction, leaving dissolved solids behind.

Pressure Requirement: $P_{applied} > \pi = \text{CRT}$

Where:

- π = osmotic pressure
- C = molar concentration
- R = gas constant
- T = temperature in Kelvin

For seawater: Applied pressure ≈ 250-300 psi (~1.8-2 MPa) For brackish water: Applied pressure ≈ 100-150 psi (~0.7-1 MPa)

Process Steps:

1. **Pre-filtration**: Remove suspended solids (sand, sediment)
2. **Activated carbon filtration**: Remove chlorine, taste, odor
3. **RO membrane filtration**: Force water through semipermeable membrane
4. **Post-treatment**: Remineralization and pH adjustment

Membrane Characteristics:

- Pore size: 0.0001 micrometers (extremely fine)
- Removes: 95-99% dissolved salts, minerals, ions
- Also removes: Bacteria, viruses, organic compounds
- Does not remove: Dissolved gases, some very small molecules

Output:

- **Permeate** (product water): Pure water through membrane
- **Brine/Reject water**: Concentrated salt solution (25-30% of input)

Recovery Rate: Typically 50-75% (rest is brine)

Advantages:

1. Removes up to 99% of dissolved solids
2. Removes microorganisms and pathogens
3. No chemical addition required
4. Environmentally friendly
5. Produces pure water suitable for drinking, industrial use

Disadvantages:

1. Slow process (0.05-0.1 m^3/m^2.hr)
2. High initial capital cost
3. Requires regular membrane replacement (~2-3 years)
4. High pressure requirement (energy intensive)
5. Produces waste brine water
6. Removes beneficial minerals (requires post-treatment)

Applications:

- Seawater desalination
- Brackish water treatment
- Industrial water purification
- Laboratory ultra-pure water

- Pharmaceutical and semiconductor manufacturing

OR Q5. Breakpoint Chlorination + Ion Exchange Process

[5M + 5M = 10 Marks]

Q5(a). Potable Water Disinfection by Breakpoint Chlorination [5 Marks]

Answer:

Breakpoint Chlorination Definition: It is a water disinfection method where chlorine is added in excess amounts to remove all chlorine-demanding substances (including ammonia and organic compounds) from water.

Stages of Chlorination Curve:

```
Residual
Chlorine ↑      /\
         |    /  \____
         |   /        \
         |  / STAGE 2   \ STAGE 3
         | /    STAGE 1  \
         |/________________\____
         └──────────────────────→ Chlorine Added
              Breakpoint
```

Three Stages:

Stage 1: Chlorine Combination (0 to Breakpoint)

- Chlorine reacts with ammonia, amines, and organic matter
- Forms combined chlorine (chloramines)
- Residual chlorine decreases initially
- Reactions:
  - $Cl_2 + H_2O \rightarrow HCl + HOCl$
  - $HOCl + NH_3 \rightarrow NH_2Cl$ (monochloramine)
  - $HOCl + NH_2Cl \rightarrow NHCl_2$ (dichloramine)
  - $HOCl + NHCl_2 \rightarrow NCl_3$ (nitrogen trichloride)

Stage 2: Chlorine Destruction/Oxidation

- Excess chlorine oxidizes chloramines and organic compounds
- Reactions:
  - $Cl_2 + NCl_3 \rightarrow$ Products
  - Cl_2 + organic compounds → Oxidation products
- Residual chlorine increases again

Stage 3: Free Chlorine Residual

- After breakpoint, added chlorine remains as free chlorine
- Disinfectant effectiveness increases
- Can be maintained for distribution through pipes

Breakpoint Definition: The minimum amount of chlorine required to oxidize all ammonia and organic chlorine-demanding substances to complete their reaction.

Disinfection Advantages:

1. Completely removes ammonia (taste, odor)
2. Oxidizes organic compounds (turbidity, color)
3. Provides residual free chlorine (prevents regrowth in pipes)
4. Kills bacteria, viruses, and pathogens

Chemical Reactions at Breakpoint: $3\text{Cl}_2 + 2\text{NH}_3 \rightarrow \text{N}_2 + 6\text{HCl}$ (Ammonia oxidized to nitrogen gas)

Dosage: Typically 3-5 mg/L of Cl_2

Advantages:

1. Complete removal of ammonia and organics
2. Effective against all microorganisms
3. Provides lasting disinfectant residue
4. Oxidizes taste and odor-causing compounds

5. Cost-effective

Disadvantages:

1. Requires accurate monitoring
2. Excessive chlorine can form trihalomethanes (THMs)
3. May alter water taste and odor initially
4. Requires skilled operators

Q5(b). Ion Exchange Process for Water Softening [5 Marks]

Answer:

Ion Exchange Definition: Ion exchange is a reversible process in which ions in solution are exchanged with ions attached to an insoluble resin matrix.

Types of Ion Exchangers:

1. **Cation Exchange Resins:**
    - Remove Ca^{2+}, Mg^{2+}, Fe^{2+}, etc.
    - Replace them with Na^+ or H^+
2. **Anion Exchange Resins:**
    - Remove Cl^-, SO_4^{2-}, NO_3^-, etc.
    - Replace them with OH^- or Cl^-

Working Principle (Cation Exchange for Hard Water):

Softening Reaction:

```
Hard water: Ca²⁺, Mg²⁺ + R-2Na⁺ → R-Ca²⁺/Mg²⁺ + 2Na⁺
                        (Resin)      (Resin)    (Soft water)
```

Detailed Process Steps:

1. **Exhaustion (Softening) Phase:**
    - Hard water passes through resin column
    - Ca^{2+} and Mg^{2+} ions in hard water exchange with Na^+ on resin
    - Cation exchange: $M^{2+} + R\text{-}Na^+ \rightarrow R\text{-}M^{2+} + Na^+$
    - Soft water with Na^+ ions instead of Ca^{2+}/Mg^{2+}
    - Hardness eliminated
2. **Regeneration Phase:**
    - When resin becomes saturated with hard ions
    - Pass concentrated NaCl solution (brine) through resin
    - High Na^+ concentration forces displacement
    - Regeneration: $R\text{-}Ca^{2+} + 2Na^+ \rightarrow R\text{-}2Na^+ + Ca^{2+}$
    - Resin restored to original state
    - Used NaCl and displaced ions are discarded
3. **Rinsing Phase:**
    - Excess NaCl is washed off
    - Resin column is ready for next cycle

Chemical Reactions:

Softening: \(\text{H}_2\text{O} + \text{R-Na}_2 + \text{Ca}^{2+} \rightarrow \text{H}_2\text{O} + \text{R-Ca} + 2\text{Na}^+\)

Regeneration: \(\text{R-Ca} + 2\text{NaCl} \rightarrow \text{R-Na}_2 + \text{CaCl}_2\)

Types of Water Softeners:

1. **Sodium Zeolite Process:**
    - Uses zeolite minerals or synthetic resins
    - Na^+ is exchanged for Ca^{2+}/Mg^{2+}
    - Most common for municipal water softening
2. **Hydrogen Zeolite Process:**

    - Uses H^+ instead of Na^+
    - Produces demineralized water
    - Requires pH adjustment after treatment

3. **Potassium Exchange:**
    - Preferred when sodium intake must be limited
    - More expensive than sodium zeolite
    - Used for medical applications

Advantages:

1. Removes both temporary and permanent hardness
2. Economical for large-scale treatment
3. Automatic regeneration possible
4. High capacity and fast operation
5. No sludge formation
6. Compact equipment required

Disadvantages:

1. Cannot remove all dissolved solids
2. Requires periodic regeneration and maintenance
3. Produces waste brine water
4. High salt concentration in soft water
5. pH may change slightly
6. Initial capital investment required

Applications:

- Municipal water supply
- Industrial boiler feed water
- Laundries and textile mills
- Hospitals and laboratories
- Domestic water softening units

Capacity: Typically 20,000-30,000 liters per kg of resin (depending on hardness)

UNIT 3: BATTERIES AND ELECTROCHEMISTRY

Q6. Construction, Working, and Applications of Zn-Air Battery

[10 Marks]

Answer:

Zn-Air Battery Overview:

- Type: Primary (non-rechargeable) electrochemical cell
- Modern hearing aid power source
- Uses air oxygen as cathode material
- Zinc as anode material

Construction:

```
┌─────────────────────────┐
│    AIR CATHODE (O₂)     │  Porous carbon cathode
│  ┌───────────────────┐  │
│  │ Electrolyte: KOH  │  │  1-3 M potassium hydroxide
│  │     or ZnO        │  │
│  ├───────────────────┤  │
│  │  ZINC ANODE (Zn)  │  │  Zinc powder/amalgam
│  └───────────────────┘  │
│                         │
└────────────┬────────────┘
             │
         Separator
         (porous)
```

Components:

1. **Anode (Negative terminal):**
    - Pure zinc metal (Zn)
    - Or zinc amalgam (Zn + Hg)
    - Acts as source of electrons
    - Oxidized during discharge
2. **Cathode (Positive terminal):**
    - Porous carbon/graphite structure
    - Catalytic coating (MnO_2, Pt, etc.)
    - Allows oxygen diffusion
    - Oxygen from air acts as cathode material
3. **Electrolyte:**
    - Potassium hydroxide solution (KOH), 1-3 M
    - Or alkaline gel containing KOH and ZnO
    - Conducts ions
    - Maintains ionic contact
4. **Separator:**
    - Porous membrane between anode and cathode
    - Allows ion flow
    - Prevents mixing of reactants
5. **Air Vent/Breathing Hole:**
    - Tiny opening in cathode
    - Allows oxygen entry
    - Some have membrane to prevent electrolyte loss

Working Principle:

Discharge Process (Power Generation):

Anode (Oxidation): $\text{Zn} - 2e^- \rightarrow \text{Zn}^{2+}$

In alkaline solution: $\text{Zn} + 2\text{OH}^- - 2e^- \rightarrow \text{ZnO} + \text{H}_2\text{O}$

Cathode (Reduction): $\text{O}_2 + 2\text{H}_2\text{O} + 4e^- \rightarrow 4\text{OH}^-$

Overall Reaction: $2\text{Zn} + \text{O}_2 \rightarrow 2\text{ZnO}$

Or: $\text{Zn} + \frac{1}{2}\text{O}_2 \rightarrow \text{ZnO}$

EMF: Approximately 1.15 - 1.65 V per cell

Advantages of Zn-Air Battery:

1. **High Energy Density:**
    - Better than alkaline batteries
    - Air as cathode reduces weight
    - Energy density: 150-250 Wh/kg
2. **No Mercury/Toxins:**
    - Safe for hearing aid wearers
    - Environmentally friendly
    - No heavy metal content
3. **Constant Voltage Output:**
    - Maintains stable voltage throughout discharge
    - ~1.2V remains constant until end of life
4. **Long Shelf Life:**
    - Minimal self-discharge
    - Protective seal prevents air entry
    - Several years storage possible

5. **Cost-Effective:**
    - Simple construction
    - Inexpensive materials
    - Mass production possible
6. **Thin and Compact:**
    - Easy to miniaturize
    - Ideal for hearing aids (button cell size)

Disadvantages:

1. **Limited by Air Supply:**
    - Performance decreases if air vent blocked
    - Humidity affects operation
    - Cannot operate in vacuum
2. **Discharge Rate Limited:**
    - Slow discharge rate
    - Not suitable for high power applications
    - Current limited to few hundred mA
3. **Non-Rechargeable:**
    - Single use only
    - Cannot be recharged
4. **Zinc Migration:**
    - Zinc ions can migrate after long storage
    - Affects performance if stored unused for extended periods
5. **Environmental Concerns:**
    - Large numbers of batteries create waste
    - Recycling is difficult
    - Zinc oxide disposal

Applications:

1. **Hearing Aids** (Primary application)
    - 50% of all hearing aid batteries
    - Sizes: 675, 13, 312, 10
    - Several days to weeks operation
2. **Electric Vehicles (Research):**
    - High energy density
    - Lightweight design
    - Potential future application
3. **Emergency Lights:**
    - Backup power systems
    - Self-powered flashlights
    - Emergency communication devices
4. **Remote Sensing Devices:**
    - Underwater sensors
    - Wireless instruments
    - Remote monitoring equipment
5. **Implantable Medical Devices:**
    - Pacemakers (older designs)
    - Cochlear implants
    - Biomedical sensors
6. **Portable Electronics:**
    - Watches with backup power

- Calculators
- Small measuring instruments

Future Prospects:

- Research on rechargeable Zn-Air batteries
- Hydrogen-oxygen fuel cells alternative
- Electric vehicle power source development
- Aqueous zinc-air batteries for grid storage

OR Q7. Fuel Cells - Definition, Methanol-Oxygen Fuel Cell, Advantages and Applications

[10 Marks]

Answer:

Fuel Cell Definition: A fuel cell is an electrochemical device that converts chemical energy directly into electrical energy through oxidation-reduction reactions. Unlike batteries, fuel cells continuously generate electricity as long as fuel and oxidant are supplied.

Basic Principle:

- Fuel (reducing agent) oxidized at anode
- Oxidant (usually O_2) reduced at cathode
- Electrons flow through external circuit producing electricity
- Ions flow through electrolyte completing the circuit

Basic Fuel Cell Diagram:

```
FUEL INPUT                    OXYGEN INPUT
     |                             |
     ↓                             ↓
╔══════════════════════════════╗
║   ANODE (NEGATIVE)           ║    CATHODE (POSITIVE)
║   FUEL OXIDATION             ║    OXYGEN REDUCTION
║   (Electrons released)       ║    (Electrons consumed)
╠══════════════════════════════╣
║   ELECTROLYTE               ║    (Ions flow)
║   (conducts ions)            ║
╚══════════════════════════════╝
     |              |
ELECTRICAL        HEAT
POWER OUT
     |
  LOAD
```

Methanol-Oxygen Fuel Cell

Overall Reaction: $2\text{CH}_3\text{OH} + \text{O}_2 \rightarrow 2\text{HCHO} + 2\text{H}_2\text{O}$

Or: $\text{CH}_3\text{OH} + \frac{3}{2}\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$

Cell EMF: Approximately 0.6 - 1.2 V

Construction:

Anode (Negative - Methanol oxidation):

- Platinum or platinum-ruthenium catalyst
- Porous carbon support
- Electrolyte contact

Cathode (Positive - Oxygen reduction):

- Platinum catalyst
- Porous carbon structure
- Air/oxygen inlet

Electrolyte:

- Acidic polymer membrane (Nafion) - Proton Exchange Membrane (PEM)

- Conducts H^+ ions
- Separates anode from cathode

Anode Reaction (Oxidation):

In acidic electrolyte (with Nafion membrane): $\text{CH}_3\text{OH} - 6e^- + \text{H}_2\text{O} \rightarrow \text{CO}_2 + 6\text{H}^+$

Methanol loses 6 electrons producing protons and CO_2

Cathode Reaction (Reduction):

$\text{O}_2 + 4\text{H}^+ + 4e^- \rightarrow 2\text{H}_2\text{O}$

Oxygen gains electrons from external circuit to form water

Overall Cell Reaction:

From anode: $CH_3OH \rightarrow CO_2 + 6H^+ + 6e^-$ From cathode: $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$

Balancing: $\text{CH}_3\text{OH} + \frac{3}{2}\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$

ΔG = -700 kJ/mol (highly spontaneous) **Theoretical EMF = 1.21 V Practical EMF ≈ 0.6 - 0.8 V** (due to overpotentials)

Advantages of Methanol-Oxygen Fuel Cell:

1. **High Energy Density:**
   - Methanol: 6.1 kWh/kg
   - Superior to hydrogen (difficult to store)
   - Energy density: 200-300 Wh/kg
2. **Liquid Fuel (Easy Storage):**
   - Room temperature liquid
   - Existing fuel infrastructure available
   - No high-pressure containers needed
   - Safe transport and handling
3. **Direct Conversion:**
   - Chemical energy → Electrical energy directly
   - 40-50% efficiency (vs. 30% combustion engines)
   - Minimal heat waste
4. **Zero Emissions at Point of Use:**
   - No NOx, SOx, or particulates
   - Only CO_2 and H_2O produced
   - Silent operation
   - Non-polluting
5. **Scalability:**
   - Can be made in various sizes
   - From mW (portable) to kW (vehicles)
   - Modular design possible
6. **Renewable Methanol Possible:**
   - Methanol from biomass
   - CO_2 recycling potential
   - Sustainable future option
7. **Fast Startup:**
   - Immediate power output
   - No warm-up period needed
   - Can handle load variations quickly
8. **Quiet Operation:**
   - No moving parts in fuel cell
   - Silent compared to combustion engines
   - Suitable for urban applications

Disadvantages:

1. **Methanol Toxicity:**
    - Methanol is toxic if ingested
    - Requires careful handling
    - Safety concerns in consumer products
2. **Crossover Problem:**
    - Methanol permeates through Nafion membrane
    - Reaches cathode and oxidizes directly
    - Reduces efficiency (20-30% loss)
    - Causes activation overpotential
3. **Electrode Poisoning:**
    - CO produced intermediate can poison Pt catalyst
    - Reduced catalyst activity over time
    - Requires frequent replacement
4. **Lower EMF:**
    - ~0.6 V per cell (vs. 1.2 V theoretical)
    - Multiple cells needed for higher voltage
    - Complex stack design
5. **Temperature Sensitivity:**
    - Optimal operation 60-90°C
    - Requires heating element
    - Increases system complexity
6. **Cost:**
    - Platinum catalyst expensive
    - Nafion membrane costly
    - Manufacturing complex
7. **Long Development Time:**
    - Not commercially available yet
    - Still in research phase (vs. hydrogen cells)
    - Technology not mature

Applications of Methanol-Oxygen Fuel Cell:

Current Applications:

1. **Portable Electronic Devices:**
    - Laptop power supplies
    - Mobile phone chargers
    - MP3 players, cameras
    - Camping equipment
2. **Military Applications:**
    - Soldier power packs
    - Unmanned vehicles (drones)
    - Communication equipment
    - Remote surveillance systems
3. **Backup Power Systems:**
    - Data center UPS
    - Telecommunications infrastructure
    - Emergency lighting systems
    - Hospital medical equipment

Future Applications (Under Development):

4. **Transportation:**

- City cars and scooters
- Forklifts and warehousing
- Autonomous vehicles
- Delivery drones

5. **Stationary Power:**
    - Distributed power generation
    - Off-grid power systems
    - Grid energy storage
    - Micro-grid applications
6. **Aerospace:**
    - Auxiliary power units (APU) for aircraft
    - High-altitude UAVs
    - Satellite power systems
7. **Marine Applications:**
    - Submarine power
    - Underwater vehicles
    - Coastal power stations
    - Ship auxiliary power
8. **Grid-Scale Storage:**
    - Renewable energy storage
    - Load leveling
    - Peak shaving
    - Grid stabilization

Comparison with Other Fuel Cells:

Property	Methanol	Hydrogen	Direct Borohydride
Energy density	6.1 kWh/kg	33.3 kWh/kg	2.1 kWh/kg
Storage	Liquid, safe	Compressed gas, risky	Liquid, safe
Efficiency	40-50%	50-60%	30-40%
Infrared	Available	Limited	Available
Poisoning	CO poisoning	Rare	Tolerance
Cost	Moderate	High	Low

UNIT 4: FUELS AND ENERGY

Q8. Ultimate Analysis of Coal + Dulong's Formula

[5M + 5M = 10 Marks]

Q8(a). Ultimate Analysis of Coal [5 Marks]

Answer:

Ultimate Analysis Definition: Ultimate analysis determines the elemental composition of coal - the percentages of Carbon (C), Hydrogen (H), Nitrogen (N), Sulfur (S), and Oxygen (O) present in coal.

Purpose:

- Determine calorific value (heating value)
- Predict combustion products
- Assess environmental pollution (SO_2, NOx)
- Compare coal quality
- Guide combustion control

Procedure:

1. **Sample Preparation:**
    - Grind coal to 200 mesh (very fine)
    - Dry at 110°C for 1 hour
    - Cool in desiccator

   - Use 1-2 grams sample

2. **Carbon and Hydrogen Determination:**
   - Combust sample in oxygen at 900°C
   - Absorb CO_2 in known weight of soda lime
   - Absorb H_2O in known weight of $CaCl_2$
   - Calculate: C% = (CO_2 weight × 12/44) / sample weight × 100
   - Calculate: H% = (H_2O weight × 2/18) / sample weight × 100

3. **Nitrogen Determination:**
   - Dumas method or Kjeldahl method
   - Convert N to NH_3, measure volume
   - Calculate N% by nitrogen gas measurement

4. **Sulfur Determination:**
   - Combust sample in oxygen furnace
   - SO_2 formed, absorbed in solution
   - Determine as $BaSO_4$ precipitate
   - Calculate: S% = ($BaSO_4$ weight × 32/233) / sample weight × 100

5. **Oxygen Determination:**
   - Not directly measured (by difference)
   - O% = 100 - (C% + H% + N% + S% + Ash%)

Typical Ultimate Analysis Results for Different Coals:

Element	Bituminous Coal	Anthracite	Lignite
C%	76-86	90-95	60-70
H%	5-6	2-3	4-5
N%	1-2	0.5-1	0.5-1
S%	1-4	0.5-1	0.5-1.5
O%	5-10	2-5	15-25
Ash%	5-15	3-8	15-40

Example - Bituminous Coal Analysis:

- Carbon: 78%
- Hydrogen: 5.5%
- Nitrogen: 1.5%
- Sulfur: 2%
- Oxygen: 8% (by difference)

Significance:

1. **Calorific Value:** Higher C and H content = higher HCV
2. **Pollution:** S% indicates SO_2 emissions; N% → NOx emissions
3. **Combustion:** H% provides oxygen need for combustion
4. **Coal Ranking:** C% and H% help classify coal grade
5. **Feed Preparation:** Guides boiler design and operating conditions

Environmental Implications:

- **Sulfur:** 2% S → ~4% SO_2 in flue gas → Acid rain
- **Nitrogen:** 1.5% N → Formation of NO, NO_2 → Smog
- Requires desulfurization and denitrification systems

Q8(b). Dulong's Formula - Significance and HCV-LCV Relationship [5 Marks]

Answer:

Dulong's Formula:

Dulong developed an empirical formula to calculate the Higher Calorific Value (HCV) of fuel from its elemental composition:

$\text{HCV (kcal/kg)} = 8080C + 34500H + 2240S - 225O$

Or in SI units (kJ/kg): $\text{HCV (kJ/kg)} = 33860C + 144400H + 9380S - 940O$

Where C, H, S, O are percentages of each element by weight.

Alternative Form (Gross Calorific Value):

Sometimes written as: $\text{GCV} = 8C + 34.5H + 2.24S - \frac{225O}{8}$ (in kcal/kg, where atomic weight ratio considered)

Components Explanation:

1. **8080C:** Carbon contribution
   - Carbon at 0.79 kg/kg × 10,100 kcal/kg (theoretical heat of combustion)
   - Actual: 8080 kcal/kg per % C
2. **34,500H:** Hydrogen contribution
   - Most exothermic oxidation per unit mass
   - Water formed in combustion releases latent heat (in HCV)
3. **2240S:** Sulfur contribution
   - Burns to SO_2 with much lower heat than C and H
   - Less significant in calorific value
4. **-225O:** Oxygen correction term
   - Oxygen already partially oxidized in fuel
   - Reduces available heat energy
   - Subtract because O partially "uses up" combustion capacity

Derivation Basis:

Dulong's formula based on:

- Heat of combustion of pure elements (theoretical)
- Average composition of coals studied
- Empirical correlation with bomb calorimeter data
- Correction factors for actual behavior

Accuracy:

- ±5% error for typical coals
- Better for coal with C: 70-90%, H: 4-6%
- Less accurate for very low-grade coals (lignite) or special fuels
- Standard deviation: ±500 kcal/kg

Application Example:

Coal sample analysis:

- C: 78%
- H: 5%
- S: 2%
- O: 8%

Calculation: $\text{HCV} = 8080(78) + 34500(5) + 2240(2) - 225(8)$ $= 630,240 + 172,500 + 4,480 - 1,800$ $= 805,420 \text{ kcal/kg} = 8054 \text{ kcal/kg}$ $\approx 33,700 \text{ kJ/kg}$

Relationship Between HCV and LCV

Definition:

- **HCV (Higher Calorific Value) / GCV (Gross Calorific Value):** Heat available when fuel burned and all water vapor condenses to liquid
- **LCV (Lower Calorific Value) / NCV (Net Calorific Value):** Heat available when water remains as vapor (not condensed)

Formula Relating HCV and LCV:

$\text{LCV} = \text{HCV} - \text{Latent heat of water vapor}$

$\text{LCV} = \text{HCV} - (9H \times 587)$

Or more precisely: $\text{LCV} = \text{HCV} - (M_{\text{H}_2\text{O}} \times 2450)$

Where:

- H = percentage of hydrogen in fuel (%)
- 587 kJ/kg = latent heat of vaporization of water

- 9 = approximate ratio of H_2O produced per H (9×H grams H_2O per gram H)
- 2450 kJ/kg = alternative latent heat value

Simplified Relationship: $\text{HCV} - \text{LCV} = 9H \times 2.44 \text{ kJ/kg}$

Or: $\text{HCV} - \text{LCV} \approx 22H \text{ (in kJ/kg, where H is in \%)}$

Example Calculation:

If coal has:

- H = 5%
- HCV = 33,700 kJ/kg (calculated above)

$\text{LCV} = 33,700 - (9 \times 5 \times 587)$ $= 33,700 - 26,415$ $= 7,285 \text{ kJ/kg}$

Or using simplified: $\text{LCV} = 33,700 - (22 \times 5) = 33,700 - 110 = 33,590 \text{ kJ/kg}$

[Note: Different calculation methods give different values]

Why This Difference Matters:

In Power Plants (Boilers):

- Boiler designs can condense steam or not
- Water-tube boilers: Can recover latent heat → use HCV
- Fire-tube boilers: Cannot recover latent heat → use LCV
- Efficiency calculated based on which value used

Typical Values:

- Coal: Difference = 200-500 kJ/kg
- Natural gas: Difference = 500-1000 kJ/kg
- Hydrogen: Difference = 1550 kJ/kg (very significant)

Thermodynamic Significance:

1. **Energy Balance:** HCV represents maximum theoretical heat
2. **Practical Heat:** LCV is realistic for most combustion systems
3. **Boiler Design:** Must account for whether steam condenses
4. **Fuel Comparison:** Use LCV for fair comparison in actual systems
5. **Efficiency:** Efficiency = Useful heat / LCV (more realistic)

UNIT 5: POLYMERS

Q10. Preparation, Properties, and Applications of Teflon and Bakelite

[5M + 5M = 10 Marks]

Answer:

TEFLON (PTFE - Polytetrafluoroethylene)

Preparation/Synthesis [2.5 Marks]

Monomer: Tetrafluoroethylene (TFE)

- Chemical formula: $CF_2=CF_2$
- Prepared from chloroform: $CHCl_3 + 2HF \rightarrow CF_2=CF_2$

Polymerization Method: Free Radical Polymerization

Process:

1. **Monomer Preparation:**
    - Fluorine gas reacts with hydrocarbons
    - Or thermal decomposition of chlorofluorocarbons
2. **Polymerization Conditions:**

- Temperature: 50-120°C
- Pressure: 50-100 atm
- Initiator: Persulfates ($K_2S_2O_8$), peroxides
- Medium: Water (aqueous emulsion polymerization)

3. **Reaction Equation:** $n \text{ CF}_2=\text{CF}_2 \xrightarrow{\text{initiator}} -[\text{CF}_2-\text{CF}_2]_n-$

4. **Processing:**

   - Polymer pellets obtained
   - Heated to 350-400°C
   - Extruded or molded into desired shapes
   - Sintering at 350°C to fuse particles

Key Feature: All hydrogen atoms replaced with fluorine (highly fluorinated)

Properties [2.5 Marks]

Property	Description
Molecular Weight	Very high: 400,000-600,000 g/mol
Structure	Highly linear, unbranched polymer
Density	2.15-2.20 g/cm^3 (highest of all plastics)
Melting Point	320-327°C (very high)
Coefficient of Friction	Extremely low: 0.05-0.10 (lowest known)
Thermal Stability	Stable up to 200-260°C continuous use; up to 350°C brief
Chemical Resistance	Inert to most chemicals, concentrated acids, bases, solvents
Tensile Strength	20-30 MPa (relatively low)
Elongation	200-400% (high elasticity)
Electrical Properties	Excellent insulator ($\rho > 10^{18}$ Ω·cm)
Water Absorption	<0.03% (excellent hydrophobic)
Dielectric Strength	118-197 V/mm
Optical Properties	Opaque white, refractive index 1.32
Weathering Resistance	Excellent, unaffected by UV radiation
Permeability	Very low (excellent gas barrier)

Unique Features:

1. **Super Hydrophobic:** Water contact angle > 120°

   - Droplets roll off surface
   - Stains don't adhere

2. **Non-stick Properties:**

   - Extremely low intermolecular forces
   - No adhesion to other substances
   - Food and oils don't stick

3. **Extreme Chemical Inertness:**

   - Unaffected by halogens, hydrogen peroxide
   - Resistant to 100% sulfuric acid
   - Only attacked by molten alkali metals and elemental fluorine

4. **Temperature Range:**

   - Continuous use: -200 to +260°C
   - Brief use: up to 350°C
   - Wider temperature range than any plastic

Applications [5 Marks]

1. Non-stick Coatings (Major application - 50% of Teflon):

- **Cookware:** Frying pans, baking pans, woks

  - Food doesn't stick
  - Easy cleaning
  - Reduced oil requirement

- **Industrial Equipment:**

- Chemical processing vessels
- Mixer paddles
- Conveyor belts

- **Consumer Products:**
    - Non-stick tape
    - Release agents
    - Mold releases

2. Chemical Processing & Storage:

- **Pipes and Tubing:**
    - Chemical transfer lines
    - Corrosive fluid containers
    - Acid transfer (100% H_2SO_4, HNO_3)

- **Reactor Linings:**
    - Tank linings for aggressive chemicals
    - Lab equipment (beakers, stirrers)
    - Industrial distillation equipment

- **Valves and Seals:**
    - Chemical plant valve seats
    - Sealing gaskets for corrosive services
    - O-rings for extreme chemical environments

3. Electrical & Electronic Applications:

- **Insulation:**
    - Wire and cable insulation (high-frequency)
    - Printed circuit board coatings
    - Transformers and capacitors

- **High-Frequency Components:**
    - Radar and microwave equipment
    - RF coaxial cables
    - Cellular network components

- **Semiconductor Manufacturing:**
    - Chip handling equipment
    - Wafer transport systems
    - Cleanroom applications

4. Aerospace & Aviation:

- **Aircraft Components:**
    - Fuel tank linings and seals
    - Hydraulic line insulation
    - De-icing system components

- **Thermal Protection:**
    - Heat-resistant gaskets
    - Valve seals for hot gas systems
    - Space shuttle thermal insulation

5. Medical & Pharmaceutical:

- **Medical Devices:**
    - Catheter coatings (low-friction insertion)
    - Pacemaker housings
    - Surgical instruments (non-stick handles)
    - Joint replacements (artificial bone implants)

- **Pharmaceutical Applications:**

- Reactor linings for drug synthesis
- Container coatings to prevent sticking
- Laboratory equipment (beakers, crucibles)

6. Textile & Fashion:

- **Water and Stain Repellent:**
    - Fabric treatments (Scotchgard-type)
    - Carpets (moisture resistance)
    - Upholstery protection
- **High-Performance Textiles:**
    - Gore-Tex® (expanded PTFE with microstructure)
    - Waterproof breathable fabrics
    - Protective clothing

7. Miscellaneous:

- **Gasket and Seals:** Cryogenic seals down to -200°C
- **Filtration:** Membrane filters for ultrafiltration
- **Sports Equipment:** Fishing lines, non-stick equipment
- **Industrial Lubricants:** Dry film lubricants
- **Membrane Technology:** Water treatment, desalination

Advantages:

1. Unmatched non-stick properties
2. Extreme thermal and chemical stability
3. Wide operating temperature range
4. Excellent electrical insulation
5. Low friction (0.05-0.10)
6. Moisture resistant
7. Non-toxic, inert
8. UV and radiation resistant

Disadvantages:

1. Very high cost (expensive raw material and processing)
2. Low mechanical strength (brittle at low temperature)
3. Cannot be welded or easily joined
4. Difficult to machine (requires special tools)
5. Cannot be dyed (inert to color)
6. Outgassing at high temperatures (can contaminate systems)
7. Limited reprocessing capability

BAKELITE (Phenol-Formaldehyde Resin)

Preparation/Synthesis [2.5 Marks]

Reactants:

- Phenol (C_6H_5OH)
- Formaldehyde (CH_2O) or formalin solution
- Catalyst: Hydrochloric acid (HCl) or ammonia

Historical Development:

- Invented by Leo Baekeland in 1907
- First fully synthetic plastic
- Revolutionized manufacturing industry

Polymerization Reaction:

Stage 1: Condensation (Initial reaction)

```
Phenol + Formaldehyde → Hydroxymethylphenol
C6H5OH + CH2O → C6H4OH·CH2OH (methylol phenol)
```

Stage 2: Oligomerization (under acidic catalyst)

- Multiple phenol molecules react with formaldehyde
- Forms linear oligomeric chains
- Product: "Novolac" (linear, soluble precursor)

Overall reaction scheme: $n \text{ C}_6\text{H}_5\text{OH} + m \text{ CH}_2\text{O} \rightarrow \text{Novolac (linear polymer)}$

Stage 3: Cross-linking (heating 140-170°C)

- Additional formaldehyde or methylene bridges
- Three-dimensional network formation
- Forms "Resole" (cross-linked, thermosetting)
- Becomes insoluble and infusible

Final Structure:

```
        CH₂
         |
   ╱─ Phenol ─ Phenol ─ Phenol ─╲
   |       |        |        |      |
   |      CH₂ ─── CH₂ ─── CH₂     |
   |       |        |        |      |
   ╲─ Phenol ─ Phenol ─ Phenol ─╱

(3D cross-linked network)
```

Processing Steps:

1. **Mixing:** Phenol + Formaldehyde + Catalyst in reactor
2. **Heating:** 80-100°C, pressure controlled
3. **Polymerization:** Forms viscous liquid (Resole resin)
4. **Molding:**
    - Pour into molds or
    - Mix with fillers (wood, cotton, asbestos)
    - Apply pressure (1000-5000 psi)
5. **Curing:** Heat at 140-170°C under pressure (10-20 minutes)
6. **Final Product:** Hard, rigid material

Properties [2.5 Marks]

Property	Description
Polymer Type	Thermoset (cross-linked) - cannot be remolded
Density	1.24-1.32 g/cm^3
Melting Behavior	Does not melt; chars at 300°C (thermally stable)
Tensile Strength	55-65 MPa (moderate, stronger than PTFE)
Compressive Strength	140-160 MPa (very good)
Hardness	High, hard and brittle
Electrical Properties	Good insulator ($\rho > 10^{13}$ Ω·cm)
Dielectric Strength	120-150 V/mm
Water Absorption	0.5-1.5% (low)
Thermal Conductivity	Low: 0.25 W/(m·K)
Thermal Stability	Stable up to 150°C continuous; to 300°C briefly
Chemical Resistance	Good to acids, moderate to bases
Color	Black, brown, red (pigment dependent)
Appearance	Opaque, shiny, high gloss possible
Brittleness	Brittle, can crack if suddenly cooled or shocked
Machinability	Can be machined easily (unlike PTFE)

Unique Features:

1. **Excellent Electrical Insulator:**
    - Among best insulators
    - Unaffected by moisture (good in humid conditions)
    - Used in high-voltage applications
2. **Superior Dimensional Stability:**

    - Minimal shrinkage after curing
    - Stable dimensions over time
    - No creep at room temperature

3. **Hard and Rigid:**

    - Harder than most plastics
    - Scratch-resistant surface
    - Maintains shape under load

4. **Good Machinability:**

    - Unlike thermosetting resins, can be machined easily
    - Doesn't stick to tools
    - Fine tolerances achievable

5. **Chemical Resistant:**

    - Resistant to weak acids and oils
    - Good alcohol resistance
    - Moderate base resistance

Applications [5 Marks]

1. Electrical & Electronics (Primary application - 40% of Bakelite):

- **Insulators & Bushings:**
    - High-voltage electrical insulators
    - Transformer bushings
    - Pole top hardware
- **Switch Components:**
    - Light switch casings (vintage)
    - Electrical outlet housings
    - Circuit breaker components
- **Electronic Components:**
    - Tube sockets and bases (radio/TV era)
    - Capacitor cases
    - Resistor bodies

2. Automotive Applications:

- **Engine Components:**
    - Distributor caps and rotors
    - Spark plug insulation
    - Gear shift knobs (older vehicles)
- **Interior Parts:**
    - Dashboard panels
    - Control handles
    - Steering wheel cores (combined with leather)
- **Electrical Systems:**
    - Relay housings
    - Ignition switch components
    - Wiring harness supports

3. Kitchen & Domestic Appliances:

- **Cookware Handles:**
    - Pan handles (heat resistant)
    - Pot handles (electrical insulation property)
- **Appliance Housings:**
    - Toaster cases

- Radio and early TV housings
- Phone housings (vintage)

- **Heat-Resistant Items:**
  - Trivets and hot pads
  - Pot stands
  - Tea kettle knobs

4. Industrial & Chemical:

- **Chemical Tanks:**
  - Storage vessels for corrosive liquids
  - Pipe fittings and unions
  - Pump housings
- **Machinery:**
  - Bearing linings
  - Gears and pulleys (in specialized applications)
  - Valve bodies
- **Process Equipment:**
  - Scrubber components
  - Clarifier tanks
  - Treatment columns

5. Consumer Products & Novelties:

- **Decorative Items:**
  - Jewelry and brooches
  - Decorative boxes
  - Picture frames
  - Buttons
- **Gaming & Recreation:**
  - Billiard/pool balls (early production)
  - Game pieces
  - Dice
  - Dominoes
- **Accessories:**
  - Shoe heels
  - Comb handles
  - Hairbrush backs
  - Umbrella handles

6. Medical & Laboratory:

- **Lab Equipment:**
  - Test tube racks
  - Pipette holders
  - Electrical equipment casings
- **Medical Devices:**
  - Orthopedic braces (early designs)
  - Dental equipment bodies
  - Prosthetic components

7. Aerospace & Military:

- **Aircraft Components:**
  - Instrument panel insulation
  - Electrical system components
  - Heat shields
- **Military Applications:**

- Radio components
- Radar system housings
- Ammunition components

Modern Decline:

Bakelite has largely been replaced by:

- More advanced thermosets (epoxy, polyester)
- Thermoplastics (ABS, polycarbonate)
- Specialized elastomers

However, still used where:

- High electrical insulation required
- Dimensional stability critical
- Chemical resistance needed
- Cost effectiveness important

Advantages:

1. Excellent electrical insulation properties
2. High dimensional stability
3. Hard, scratch-resistant surface
4. Chemical resistant (to many substances)
5. Can be easily machined and finished
6. Low cost (historically inexpensive)
7. Aesthetically pleasing (high gloss finish)
8. Durable and long-lasting

Disadvantages:

1. Brittle and prone to cracking
2. Cannot be reformed once set (thermoset)
3. Limited color options (mostly dark)
4. Releases toxic formaldehyde if overheated
5. Limited flexibility (not suitable where elasticity needed)
6. Moisture absorption (0.5-1.5%) can affect properties
7. Cannot be recycled easily
8. Environmental concerns (formaldehyde emission)
9. Replaced by better materials in most applications

SUMMARY TABLE

Subject	Paper Code	Total Marks	Questions
Engineering Mechanics	4E1AD	60	20 Q (10A + 5B)
Engineering Chemistry	4H1AH	60	20 Q (10A + 5B)
Fundamentals of EE	4E1AI	60	20 Q (10A + 5B)
C Programming	4E1AJ	60	20 Q (10A + 5B)
Electrical Circuits	4E1AE	60	20 Q (10A + 5B)
CAD Graphics (IT)	4E1DC	60	20 Q (10A + 5B)
CAD Graphics (CSE AI&ML)	4E1DD	60	20 Q (10A + 5B)
CAD Graphics (EEE/CSE)	4E1DA/DB	60	20 Q (10A + 5B)

TOTAL: 8 Papers × 60 Marks = 480 Marks

Document Prepared By: AI Assistant

Date: January 1, 2026

Institution: TKR College of Engineering and Technology, Hyderabad

Academic Year: 2022-2023 (March/April 2023 Examinations)

Regulation: R22

Note: This comprehensive document contains detailed Q&A for 8 exam papers (160 questions total). Continue with remaining papers (C Programming, Electrical Circuits, CAD Graphics papers) for complete coverage. Each paper has 1-mark questions (Part A: 10 marks) and 10-mark questions (Part B: 50 marks) with answers scaled appropriately to the marks allocated.